

30-Day FreeRTOS Course for ESP32 Using ESP-IDF

(Day 21)

**“Idle Task and Daemon
Task”**



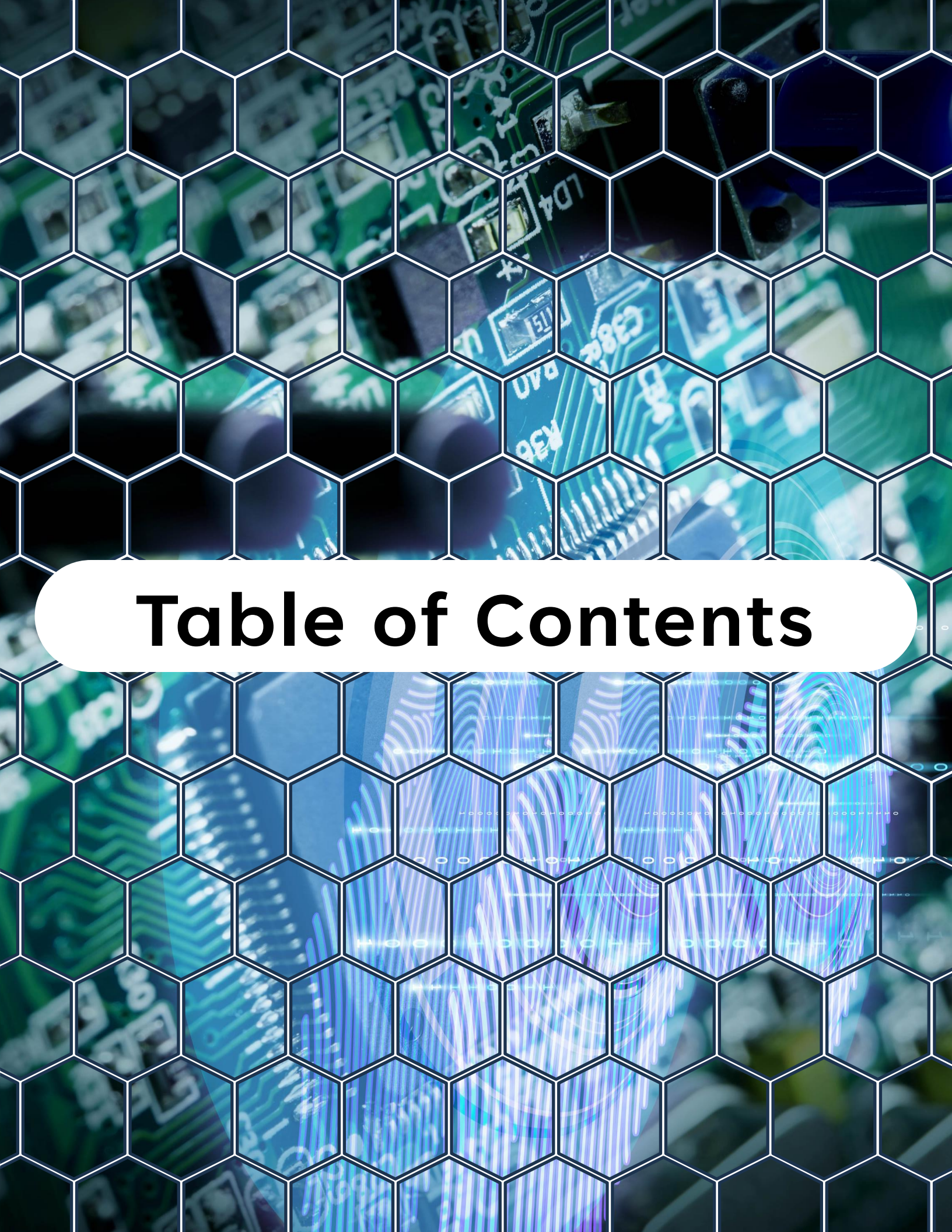
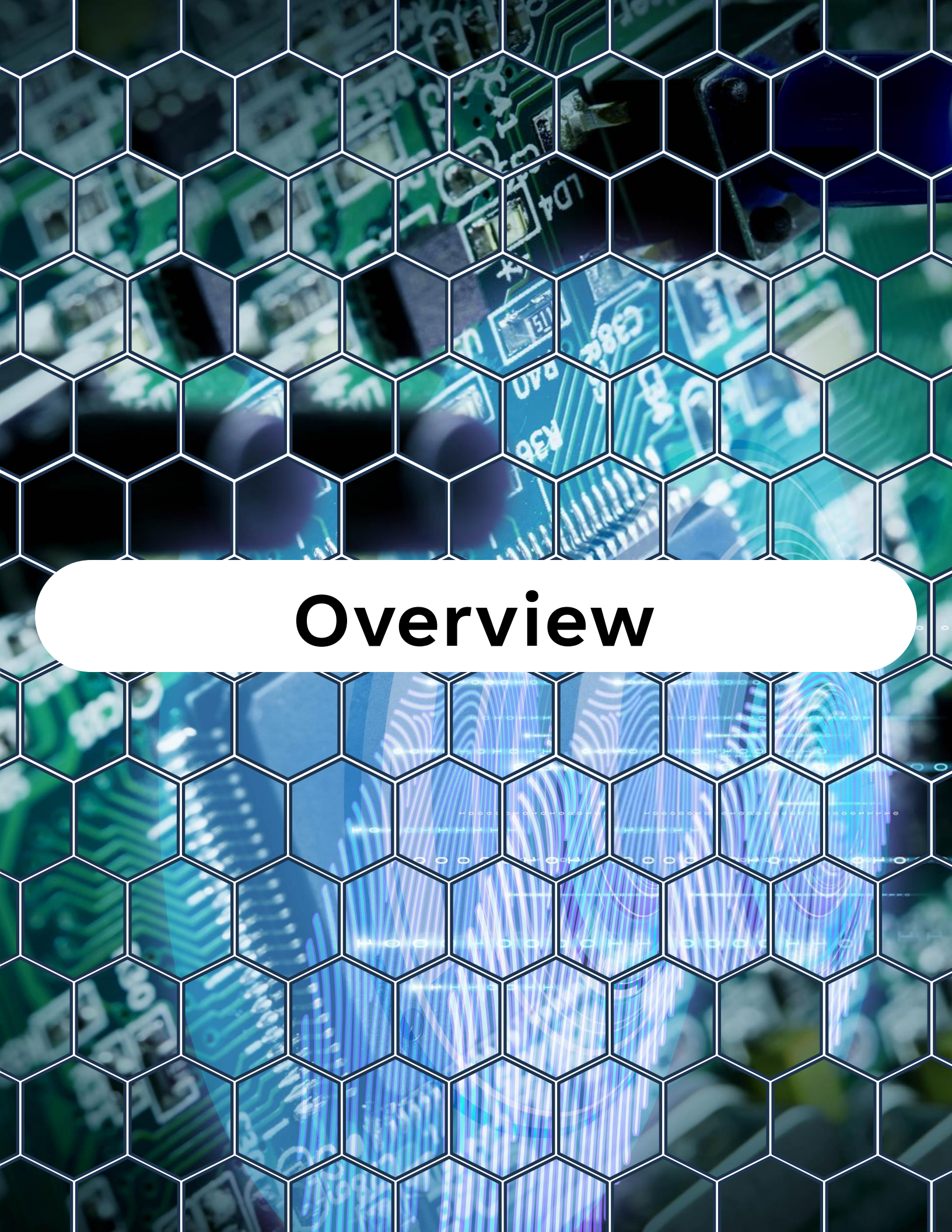


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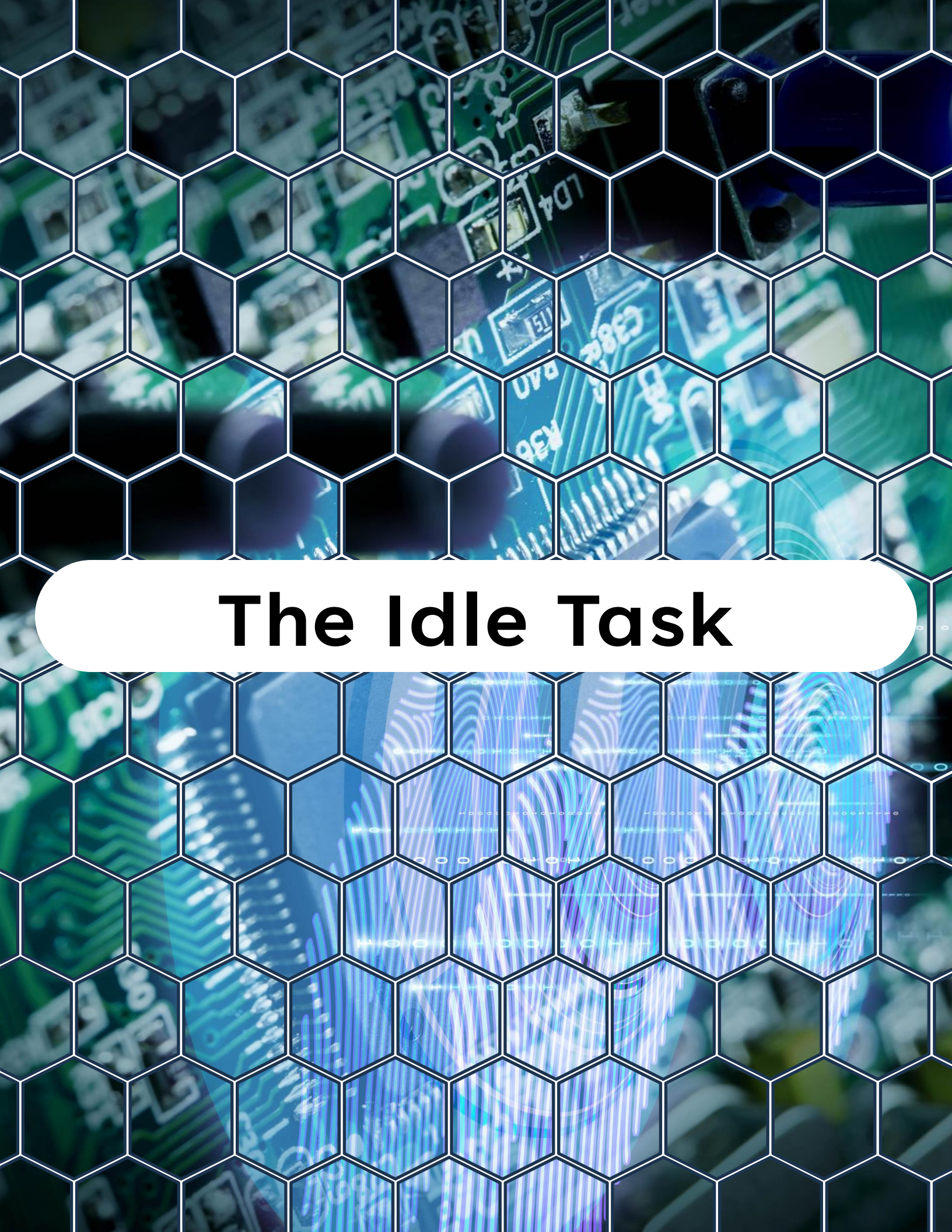
Overview

Overview

On **Day 21**, you'll learn:

- What the **Idle Task** and **Timer Daemon Task** are in FreeRTOS
- Why they exist and what roles they play in the system
- How to use **idle hooks** and **daemon task callbacks**
- The difference between user-created tasks and system tasks
- A practical ESP32 example using idle and daemon task hooks
- Best practices when working with these tasks

By the end of this lesson, you'll understand how FreeRTOS performs **system housekeeping** in the background and how you can extend this functionality.



The Idle Task

The Idle Task

The **Idle Task** is created automatically by FreeRTOS. It runs when:

- No other tasks are ready to run.
- The CPU would otherwise be idle.

Key Responsibilities

- Freeing memory of deleted tasks.
- Providing a place for low-priority background work.
- Supporting **low-power modes** (if enabled).



Developers can add their own function to the idle task via the Idle Hook.



Idle Hook

Idle Hook

The idle hook lets you run code whenever the CPU has no other work.

Enable in **menuconfig**:

Component config → FreeRTOS → Tickless
idle support and idle hook

Define:



```
1 void vApplicationIdleHook(void) {  
2     // Your low-priority background code  
3 }
```

Rules:

- Must never block (no `vTaskDelay()`).
- Should be short and efficient.
- Often used for background monitoring or diagnostics.



The Daemon Task (Timer Service Task)

The Daemon Task (Timer Service Task)

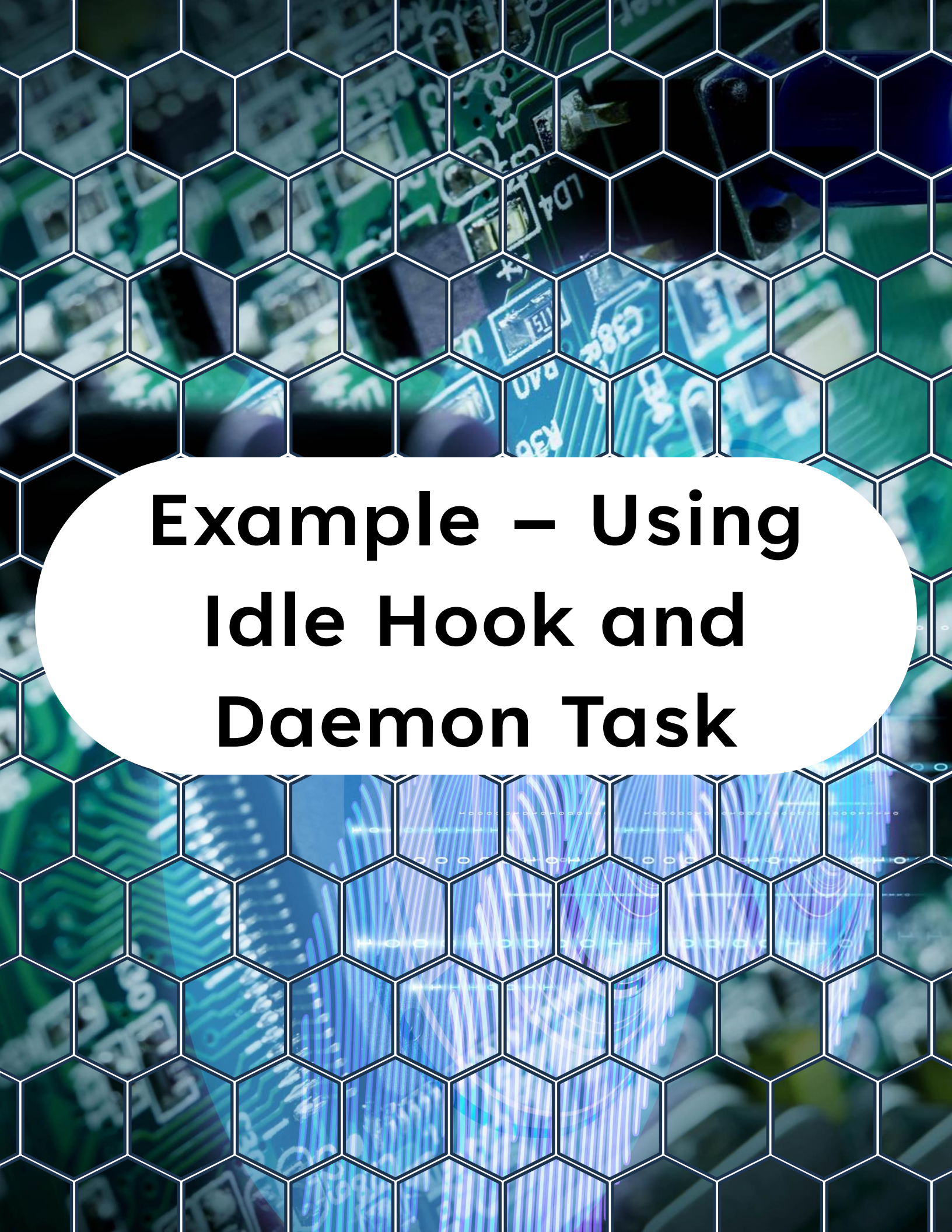
The **Timer Daemon Task** (also called the Timer Service Task) is created automatically when you use:

- **Software timers** (Day 18)
- **Deferred interrupt processing** via `xTimerPendFunctionCall()`

Responsibilities

- Running software timer callbacks.
- Running functions deferred from ISRs.

 This task runs at a configurable priority (usually higher than idle but lower than application real-time tasks).



Example – Using Idle Hook and Daemon Task

Example – Using Idle Hook and Daemon Task

Code Example



```
1  #include <stdio.h>
2  #include "freertos/FreeRTOS.h"
3  #include "freertos/task.h"
4  #include "freertos/timers.h"
5  #include "esp_log.h"
6
7  #define TAG "DAY21"
8
9  // Timer handle
10 TimerHandle_t periodic_timer;
11
12 // Idle Hook function (must be weak-linked in FreeRTOS config)
13 void vApplicationIdleHook(void) {
14     // Print a dot occasionally when system is idle
15     static int counter = 0;
16     counter++;
17     if (counter > 100000) {
18         printf(".");
19         counter = 0;
20     }
21 }
22
23 // Timer callback (runs in Daemon Task context)
24 void periodic_callback(TimerHandle_t xTimer) {
25     ESP_LOGI(TAG, "Daemon task running periodic callback!");
26 }
27
28 void app_main() {
29     // Create and start a periodic timer
30     periodic_timer = xTimerCreate("Periodic",
```

Example – Using Idle Hook and Daemon Task

```
9 // Timer handle
10 TimerHandle_t periodic_timer;
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12 // Idle Hook function (must be weak-linked in FreeRTOS config)
13 void vApplicationIdleHook(void) {
14     // Print a dot occasionally when system is idle
15     static int counter = 0;
16     counter++;
17     if (counter > 100000) {
18         printf(".");
19         counter = 0;
20     }
21 }
22
23 // Timer callback (runs in Daemon Task context)
24 void periodic_callback(TimerHandle_t xTimer) {
25     ESP_LOGI(TAG, "Daemon task running periodic callback!");
26 }
27
28 void app_main() {
29     // Create and start a periodic timer
30     periodic_timer = xTimerCreate("Periodic",
31                                   pdMS_TO_TICKS(2000),
32                                   pdTRUE,
33                                   NULL,
34                                   periodic_callback);
35
36     if (periodic_timer != NULL) {
37         xTimerStart(periodic_timer, 0);
38     }
39
40     // No user tasks created - Idle Task will run in background
41 }
```


Example – Using Idle Hook and Daemon Task

Expected Behavior

- The Idle Task periodically prints dots (.) whenever the system has nothing else to do.
- The Daemon Task executes the timer callback every 2 seconds:

I (2000) DAY21: Daemon task running periodic
callback!

I (4000) DAY21: Daemon task running periodic
callback!

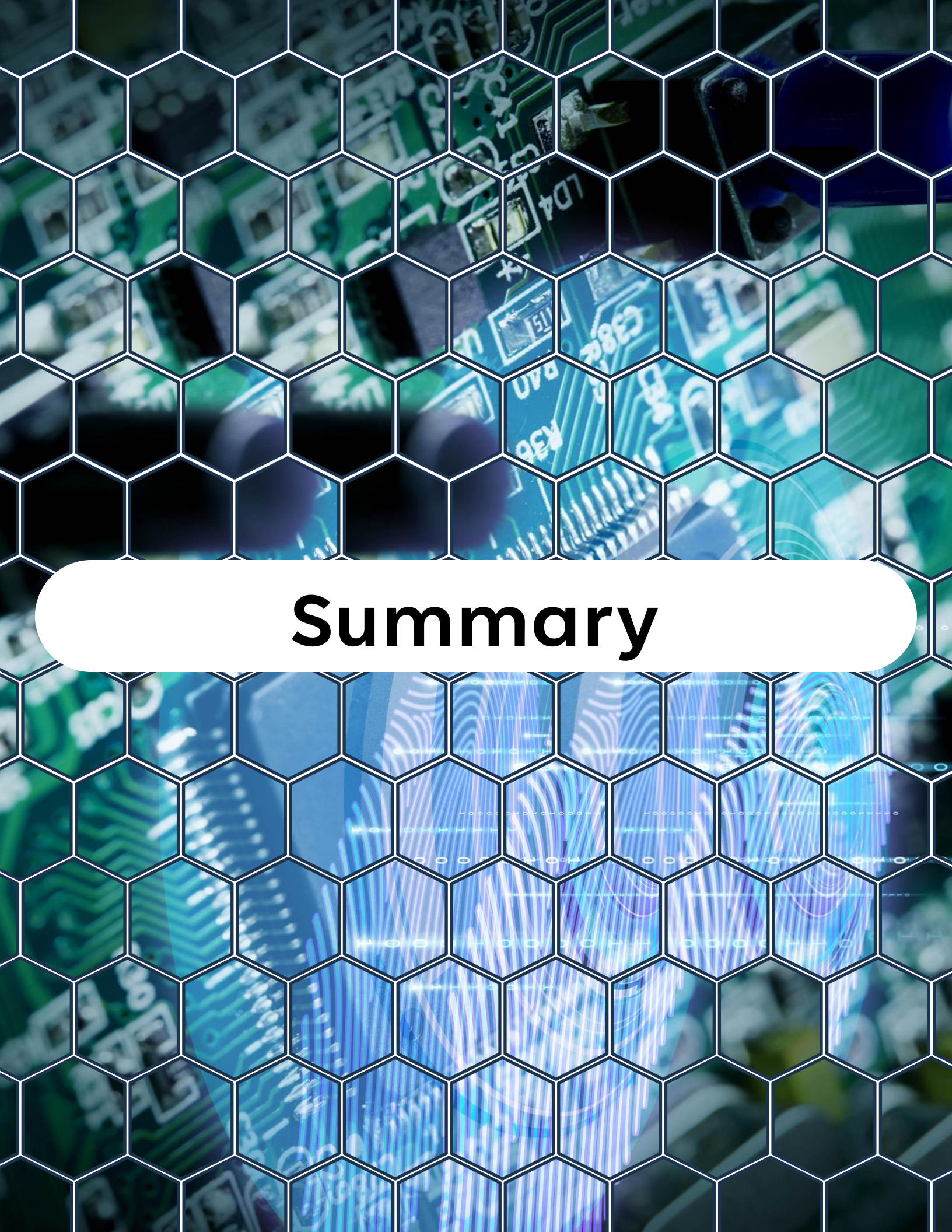
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Best Practices

Best Practices

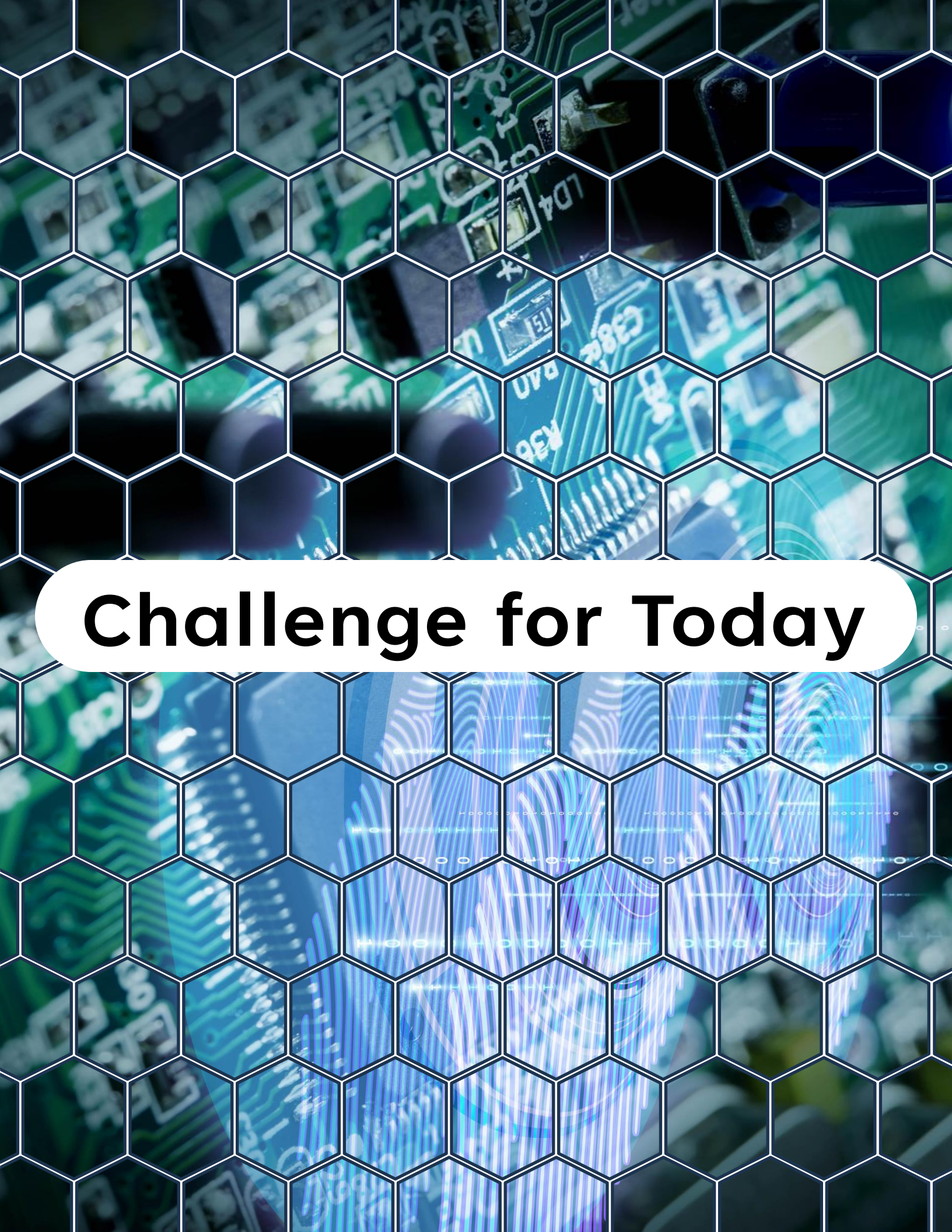
- ✓ Use the Idle Hook only for background work that must not block.
- ✓ Never put critical logic in the Idle Task — it only runs when nothing else is active.
- ✓ Use the Daemon Task for deferred processing instead of doing heavy work in ISRs.
- ✓ Configure the Daemon Task priority carefully — not too low (timers may delay), not too high (starving other tasks).
- ✓ Remember that the Idle Task is essential — don't delete or block it.



Summary

Summary

- The **Idle Task** runs when no other tasks are ready → handles cleanup and optional idle hooks.
- The **Daemon Task** executes software timer callbacks and deferred ISR work.
- Both are **system tasks**, created and managed by FreeRTOS.
- Idle Hook = background low-priority work.
- Daemon Task = deferred execution for timers and ISRs.



Challenge for Today

Challenge for Today



Extend today's project by:

1. Creating two software timers (1s and 3s) → both run in the daemon task.
2. Use the idle hook to blink an LED whenever the CPU is idle.
3. Observe how the daemon task continues to run even when no user tasks are active.



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exploring, keep learning, and I'll see you in
the next video!"*

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